HealthConnect Analytics Testing and Refinement Report

Week 7 | Data Analytics | Checking that the Week 6 work is right

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Builds on	Week 6: HealthConnect Rwanda hospital analytics (database, dashboard, report, slides)
Evidence	Test list (120 tests), code, screenshots, issue log and decision log are in the Week7 folder

1. The short story

In Week 6 I built a dashboard and reached some findings. This week I checked whether they are right. I did three things: re-calculated the numbers from the original files, tested whether each finding could just be chance, and tested the dashboard in Power BI.

- **The numbers are right, with one small mistake.** 38 of 39 numbers matched exactly. The share of poor ratings is **40.4%**, not 40.5%. I fixed it everywhere.
- **Some Week 6 findings are too weak to keep.** The differences between hospitals, departments and provinces are small enough to be chance. I removed those findings and changed the recommendations.
- **Four dashboard charts were not showing.** I fixed them in a new version of the dashboard. I still need to retest them in Power BI.

What I tested	Tests	Passed	Problem found
Original files match the Power BI files	34	34	0
Numbers re-calculated from the original files	42	39	3
Database results match a second method	8	8	0
Dashboard cards and charts	17	13	4
Slicers (filters) change every card	1	0	1
Week 6 findings	18	8	10
Total	120	102	18

2. Where this starts from

Question	Answer
Week 6 output	HealthConnect Rwanda hospital analytics: 27 datasets, 40 SQL questions, a six-page Power BI dashboard, a report and slides
Main result	RWF 9.31 billion billed, 90.1% collected, RWF 919.9 million unpaid, half of the insurance claims pending
Main limit	Only 5,000 of 60,000 appointments have hospital details, and there is no no-show field
What I tested	Whether the numbers, the dashboard and the findings are correct
Other track	Project Management (section 7)

Note: the Week 7 brief talks about clinic attendance data. My Week 6 work used the Rwanda data, and the brief says to continue from Week 6, so I continued with the Rwanda data. It has no attendance field, so I could not test attendance findings.

3. How I tested

- I re-calculated each number from the original files with separate code, so a mistake in Week 6 could not be repeated.
- I opened the dashboard in Power BI and compared every card with the expected number.
- For each finding I asked one question: **could this difference just be chance?** If yes, I do not treat it as a real finding.

4. Are the numbers right?

Yes. 38 of 39 matched, including revenue (RWF 9,309,401,961), money collected (8,389,454,556), unpaid (919,947,405), patients seen (4,443) and average rating (2.99). In Power BI, all 13 cards I could see showed the expected number.

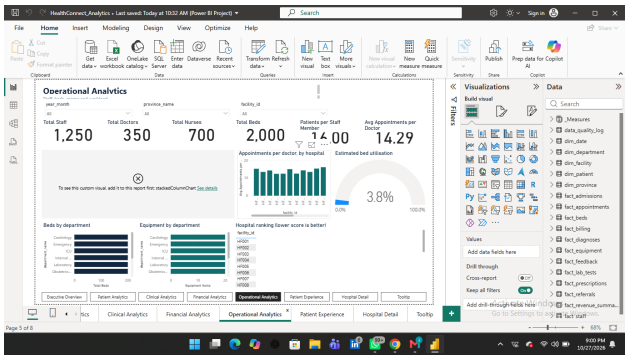
Number	Week 6	Correct	Action
Poor ratings (1 or 2 stars)	40.5%	40.4% (8,518 of 21,064)	Fixed in the report, slides, README and measures. I checked the new files: they show 40.4%

I also found two data problems that do not change any number: many patients under 18 are recorded as married or widowed, and some height and weight values are not realistic. I did not use those fields.

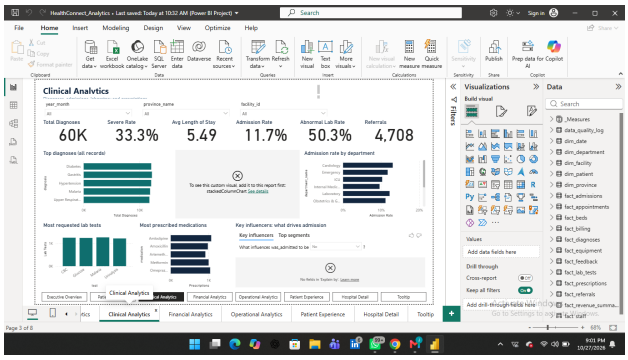
5. Dashboard problems and fixes

Problem	What I saw	Fix
Stacked column charts empty	Grey box saying "add this custom visual"	Chart type corrected in the new version
Key influencers empty	"No fields in Explain by"	Fields added in the new version
Ranking table incomplete	Only the hospital column showed	Drag the score and rank into the table (checklist R3)
Map blank	Provinces did not show	New column for province names; bar chart if still blank
Some cards ignore filters	For example Total Patients ignores the hospital filter, because patients have no hospital	Notes added on the pages

Before: Operational page



Before: Clinical page



The after screenshots go in the Evidence folder once the new version is opened in Power BI.

6. Do the Week 6 findings still hold?

Week 6 finding	Could it be chance?	Decision
The busiest hospitals are the least liked	Yes	Removed
Some hospitals are rated better or worse than others	Yes. Each hospital has only 72 to 102 ratings	Removed
ICU is the worst department	Yes	Removed
Kigali City uses hospitals least	Yes	Removed
Hypertension is becoming more common	Yes	Removed
Hospitals differ in how busy they are	Maybe (borderline)	Reworded: "similar in volume"
Workload per doctor is uneven (12 to 16 appointments)	Probably not, but the gap is small	Kept, but only to watch
RWF 919.9 million is unpaid	No. It matches in three ways	Kept
Half of the insurance claims are pending	No	Kept

Another problem: in 6,336 of 21,064 records the star rating disagrees with the written comment. So satisfaction numbers are labelled "as recorded; not confirmed".

7. Working with another track

The Project Manager, Manal El khalifi, asked on Slack whether two model results looked right: a claims model (accuracy 50.1%, about the same as a coin toss) and a triage model (finds 9.7% of emergencies).

Step	What happened
Test	I checked whether insurance claim approval depends on anything in the data, and counted emergency visits
Finding	Claims are 50% approved and 50% pending whatever the insurer, payment method, amount or visit type. So the data gives a model nothing to learn from, and about 50% accuracy is expected. There are 1,650 emergency visits, and 20% of them is 330, the number she quoted
Action	I sent her both results and asked her to confirm the claims target and the emergency labels
Reply	She thanked me and said she had submitted (2 Nov, 12:10 AM). She did not report a changed result, so there was **no retest**

The first three steps were done and acknowledged. The retest did not happen, and I have recorded it that way.

8. What I changed

Item	Before (Week 6)	After (Week 7)
Poor ratings	40.5%	40.4%
Hospital rating colours	4 hospitals red	None flagged: differences are within chance
Moving doctors between hospitals	Do it now	Watch monthly and recheck with full data
Kigali City recommendation	Look at access	Removed: no evidence
Patient experience programme	Start one for busy hospitals and ICU	First check the rating data
Dashboard	Version 1	Version 2: charts fixed, notes added, sample-size warning on small tables

9. Updated recommendations

- **Chase the RWF 919.9 million unpaid** (11,887 bills). This is the strongest finding.
- **Clear the pending insurance claims** (21,028 claims, RWF 2.44 billion).
- **Check the rating data** before running any patient-experience programme.
- **Watch doctor workload monthly**; do not move staff yet.
- **Fix the data:** add an attendance field, hospital names, the full 60,000 appointments, and a reliable bill status.

10. What is still open, and Week 8

Issue	What to do
The data has no attendance or no-show field, so the main project question cannot be answered	Confirm the right dataset with the programme (most important)
Only 5,000 of 60,000 appointments have hospital details	Ask for the full list and rerun the tests
Star ratings disagree with comments	Ask the data owner which one is right
Dashboard fixes not yet retested in Power BI	Open version 2, take after screenshots, save the .pbix and PDF
Revenue Summary (RWF 1.13 billion) does not match billing (RWF 9.31 billion)	Ask the data owner; billing is used for totals

Hospitals have only 70 to 100 ratings each, so small differences cannot be told apart from chance. "Removed" means "not proven", not "proved equal".